

LAB 3

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Presented to: **Gina Scaldaferri**

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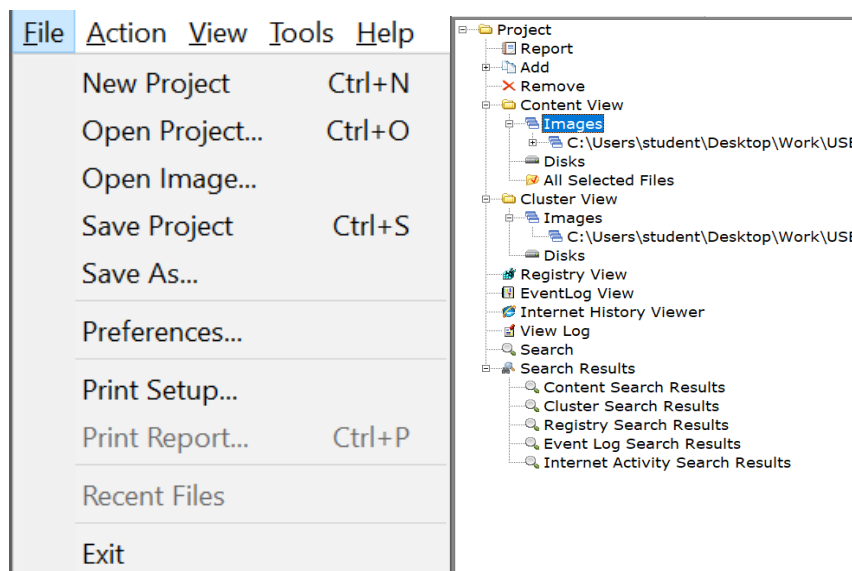
Introduction

In this lab, two hard disk drives (Drive U: and Drive F:) were acquired and converted into forensic images using ProDiscover and FTK Imager. These images were then analyzed using the same tools. The objectives of this analysis were to calculate hash values, identify the types of files present in each image, examine deleted files, and determine whether any suspicious files existed within the drives.

Analysis

1. Viewing images in Pro Discover:

To begin the analysis, ProDiscover was opened and the disk image was loaded by navigating to File → Open Image. The image file located at C:\Users\student\Desktop\Work\USB-Pro-U was selected. After loading, the image structure appeared in the left navigation pane, where multiple views such as Content View, Cluster View, Registry View, and Search Results were available for examination.



What is the hash of the image?

The following hash values were generated for the **USB-Pro-U** image:

- MD5 - 79F0B5D05C44189598FA9611FE504AEC
- SHA1 - AFD20A96C5EEDD04AEFD649A9D381172CDD5DAF4
- MD5 checksum – ad08bb6f03af12b30b0cf7571439d8e6

What files are present in the image?

The image contained multiple file types, including **DOC**, **XLS**, **PNG**, **LOG**, **INI**, and **DAT** files. Only one file was marked as deleted. The largest file in the image was **\$BadClus:\$Bad**, which is a critical NTFS metadata file used to track unusable (bad) sectors on the disk.

Select	File Name	File Ext...	Size	Attributes	Deleted	Created ...	Modified ...	Accessed ...	Parent Fo...	SHA1 C
☐	\$Extend			---META---	NO	08/02/2...	08/02/2...	08/02/2...	C:\User...	
☐	\$RECYCLE.BIN			-----...	NO	08/02/2...	08/02/2...	08/05/2...	C:\User...	
☐	System Volume Inf...			-----...	NO	08/02/2...	08/02/2...	09/09/2...	C:\User...	
☐	Deleted Files			-----...	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	All Files			-d-----	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	\$AttrDef		0 bytes	---META---	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	\$BadClus		0 bytes	---META---	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	\$BadClus:\$Bad		4,291,817,472 b...	---ADS---	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	\$Bitmap		0 bytes	---META---	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	\$Boot		0 bytes	---META---	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	\$LogFile		0 bytes	---META---	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	\$MFT		16,384 bytes	---META---	NO	08/02/2...	08/02/2...	08/02/2...	C:\User...	
☐	\$MFTMirr		0 bytes	---META---	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	\$Secure		0 bytes	---META---	NO	08/02/2...	08/02/2...	08/02/2...	C:\User...	
☐	\$Secure:\$SDS		265,304 bytes	---ADS---	NO	08/02/2...	08/02/2...	08/02/2...	C:\User...	
☐	\$UpCase		0 bytes	---META---	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	\$UpCase:\$Info		32 bytes	---ADS---	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	\$Volume		0 bytes	---META---	NO	12/31/1...	12/31/1...	12/31/1...	C:\User...	
☐	20160425_142807(0)	jpg	2,648,568 bytes	-----...	NO	08/02/2...	07/22/2...	08/05/2...	C:\User...	
☐	Annual Payroll 2	xls	32,768 bytes	-----...	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...	
☐	Balance Sheet	xls	30,720 bytes	-----...	NO	08/02/2...	07/07/2...	08/05/2...	C:\User...	
☐	Co Emp	xls	27,136 bytes	-----...	NO	08/02/2...	07/07/2...	08/05/2...	C:\User...	
☐	DCP_1255	jpg	271,844 bytes	-----...	NO	08/02/2...	09/20/2...	08/05/2...	C:\User...	
☐	Employer List	doc	33,792 bytes	-----...	NO	08/02/2...	07/07/2...	08/05/2...	C:\User...	
☐	Images Profit	xls	48,128 bytes	-----...	NO	08/02/2...	07/07/2...	08/05/2...	C:\User...	
☐	IMG_20160111_16...	png	922,357 bytes	-----...	NO	08/02/2...	12/12/2...	08/05/2...	C:\User...	
☐	IMG_20160113_15...	png	1,130,153 bytes	-----...	NO	08/02/2...	12/12/2...	08/05/2...	C:\User...	
☐	LairNetPutty	exe	516,096 bytes	-----...	NO	08/02/2...	04/26/2...	08/31/2...	C:\User...	
☐	Online	docx	16,876 bytes	-----...	NO	08/02/2...	07/07/2...	08/05/2...	C:\User...	
☐	Profit Potential	xls	28,160 bytes	-----...	NO	08/02/2...	07/07/2...	08/05/2...	C:\User...	
☐	Qtr 1 Emp	xls	23,040 bytes	-----...	NO	08/02/2...	07/07/2...	08/05/2...	C:\User...	
☐	Rocky Mountain Ou...	doc	23,040 bytes	-----...	NO	08/02/2...	07/07/2...	08/05/2...	C:\User...	
☐	Screenshot_2016-0...	png	159,777 bytes	-----...	NO	08/02/2...	05/22/2...	08/05/2...	C:\User...	
☐	Stock Club	xls	72,704 bytes	-----...	NO	08/02/2...	07/07/2...	08/05/2...	C:\User...	
☐	Summary	xls	36,864 bytes	-----...	NO	08/02/2...	07/07/2...	08/05/2...	C:\User...	

Follow the same steps to open the image USB-Pro-F:

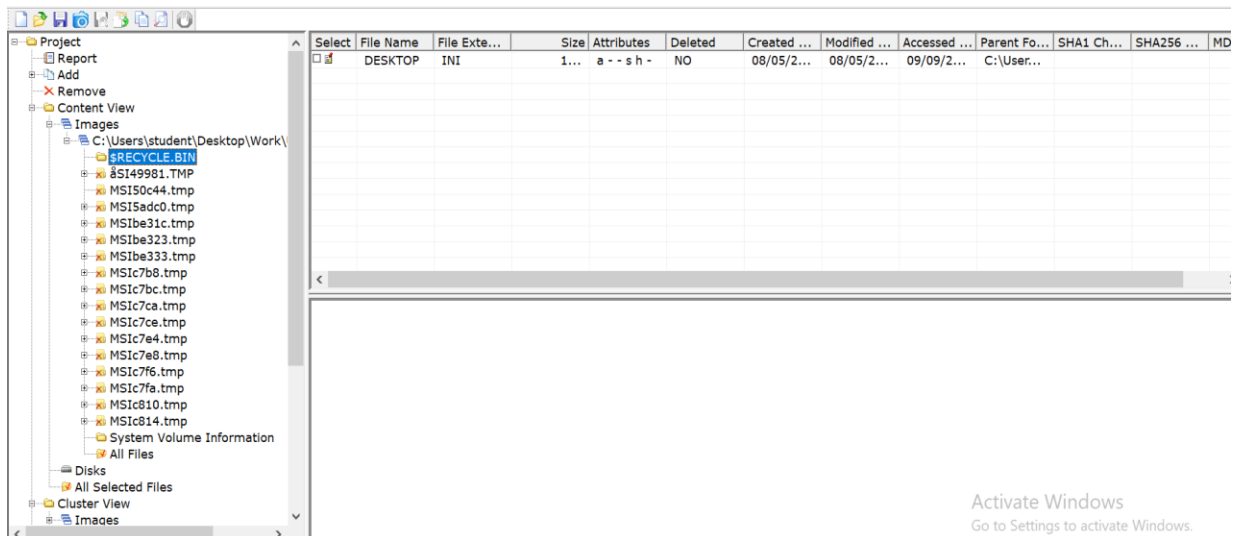
What is the hash of the image?(USB-Pro-F)

These are the hash values of image USB-Pro-F

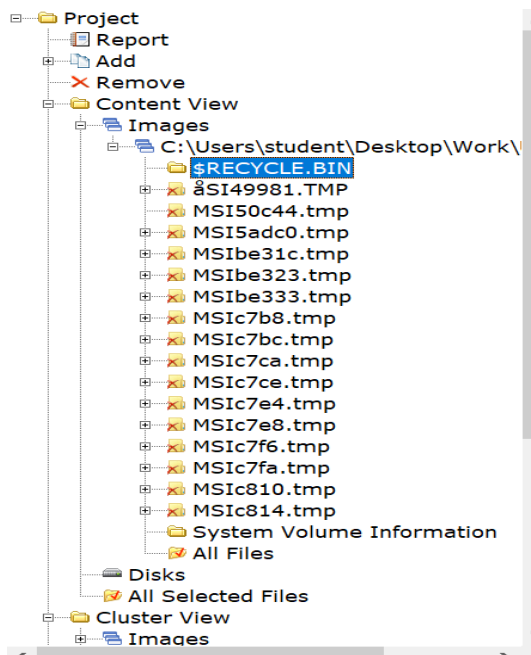
- MD5 - BBCAE44CD0834DE08271CFC5CE71BC5F
- SHA1 - 8F97BA5CC91CE7C58957456DFAD016F71D4F5603
- MD5 Checksum - cf302355dbff780e924dc247cc63e6da

What files are present in the image?

The file structure of the USB-Pro-F image was largely similar to that of USB-Pro-U. Inside the \$RECYCLE.BIN directory, a desktop.ini file was found. This file is a hidden Windows system file used to store folder display settings.



Additionally, the image contained a System Volume Information folder, which stores system restore points and related metadata. Several deleted .tmp files were present, along with file types such as PNG, DOC, XLS, JPG, and JPEG.



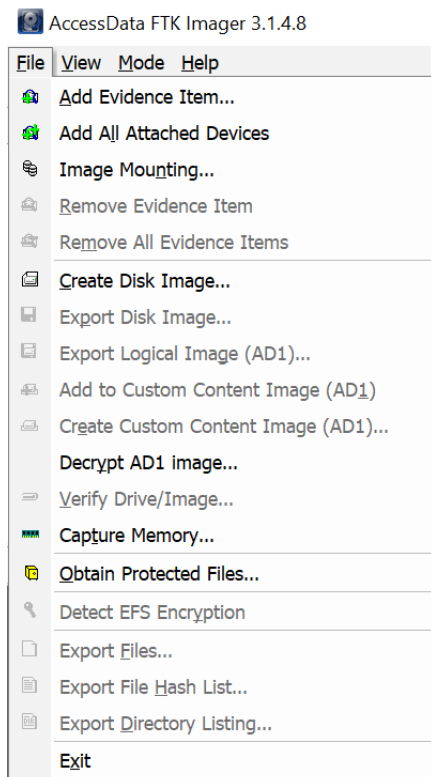
One notable file found was **LairNetPutty.exe**, which appears to be a duplicate or renamed version of the PUTTY executable. This was the only executable file identified in the image.

Select	File Name	File Ext...	Size	Attributes	Deleted	Created ...	Modified ...	Accessed ...	Parent Fo...	SHA1 Ch...	SHA256 ...	MD5
	Online	docx	16,...	a - - - -	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	Profit P...	xls	28,...	a - - - -	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	Qtr 1 Emp	xls	23,...	a - - - -	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	Rocky ...	doc	23,...	a - - - -	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	Screens...	png	159,...	a - - - -	NO	08/05/2...	05/22/2...	08/05/2...	C:\User...			
	Stock Club	xls	72,...	a - - - -	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	Summary	xls	36,...	a - - - -	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	201604...	jpg	2,648...	a - - - -	NO	08/05/2...	07/22/2...	08/05/2...	C:\User...			
	Annual ...	xls	32,...	a - - - -	YES	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	Balance...	xls	30,...	a - - - -	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	Co Emp	xls	27,...	a - - - -	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	DCP_1255	JPG	271,...	a - - - -	NO	08/05/2...	09/20/2...	08/05/2...	C:\User...			
	Employ...	doc	33,...	a - - - -	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	Images ...	xls	48,...	a - - - -	NO	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	IMG_20...	png	922,...	a - - - -	NO	08/05/2...	12/12/2...	08/05/2...	C:\User...			
	IMG_20...	png	1,130...	a - - - -	NO	08/05/2...	12/12/2...	08/05/2...	C:\User...			
	LairNet...	exe	516,...	a - - - -	NO	08/05/2...	04/26/2...	08/05/2...	C:\User...			
	Annual ...	xls	32,...	a - - - -	YES	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	Annual ...	xls	32,...	a - - - -	YES	08/05/2...	07/07/2...	08/05/2...	C:\User...			
	Indexer...		7...	a - - - -	NO	08/05/2...	08/05/2...	09/09/2...	C:\User...			
	WPSetti...	dat	1...	a - - - -	NO	08/11/2...	08/11/2...	09/09/2...	C:\User...			
	DESKTOP	INI	1...	a - - s h -	NO	08/05/2...	08/05/2...	09/09/2...	C:\User...			

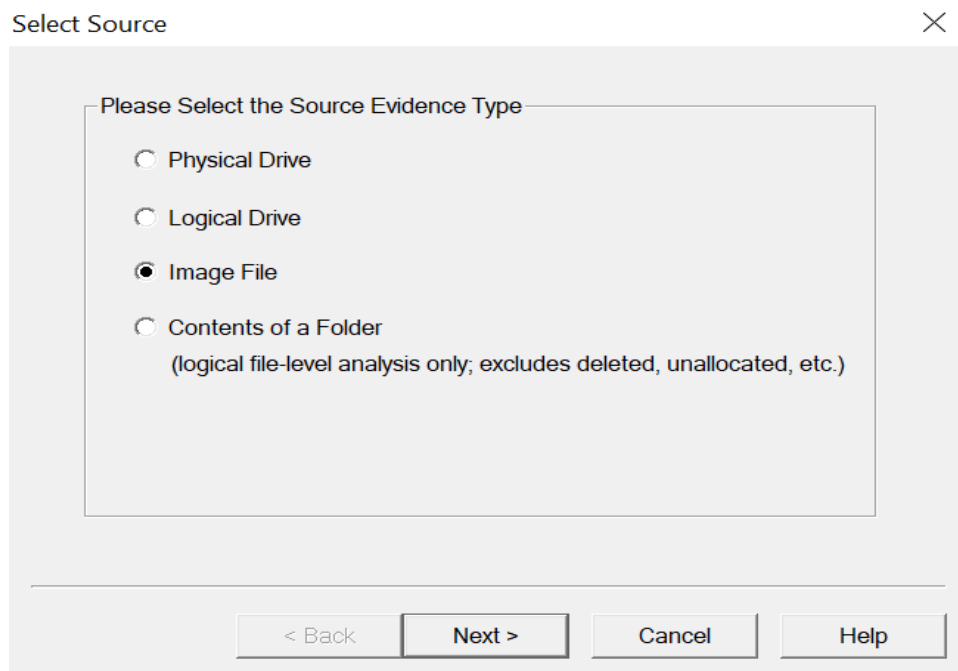
2. Viewing images in FTK Imager

Steps to add image:

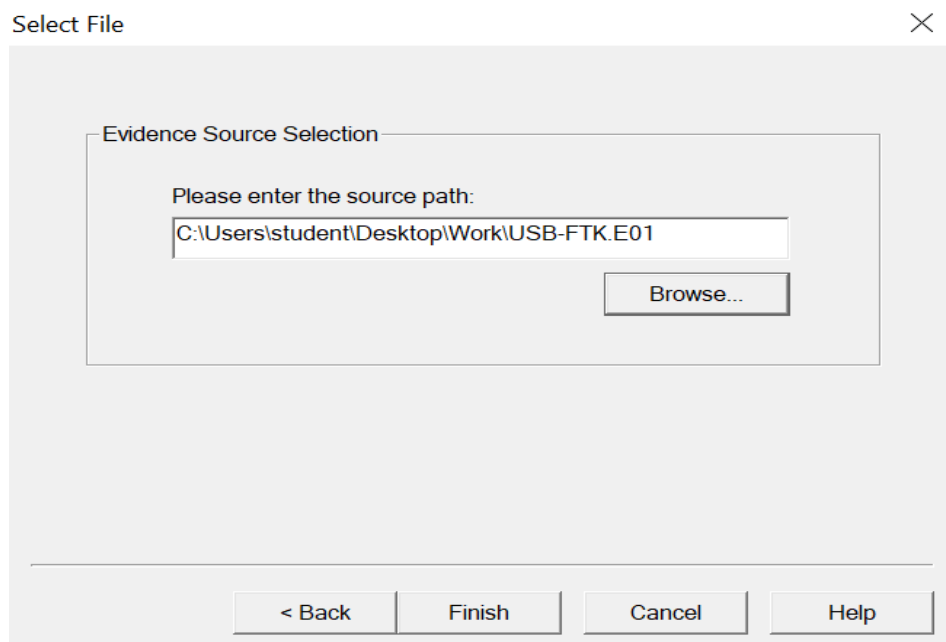
FTK Imager was launched, and the image was added by selecting **File → Add Evidence Item**. The **Image File** option was chosen, and the corresponding image file was browsed and loaded. Once added, the image contents became visible in the **Evidence Tree** pane.



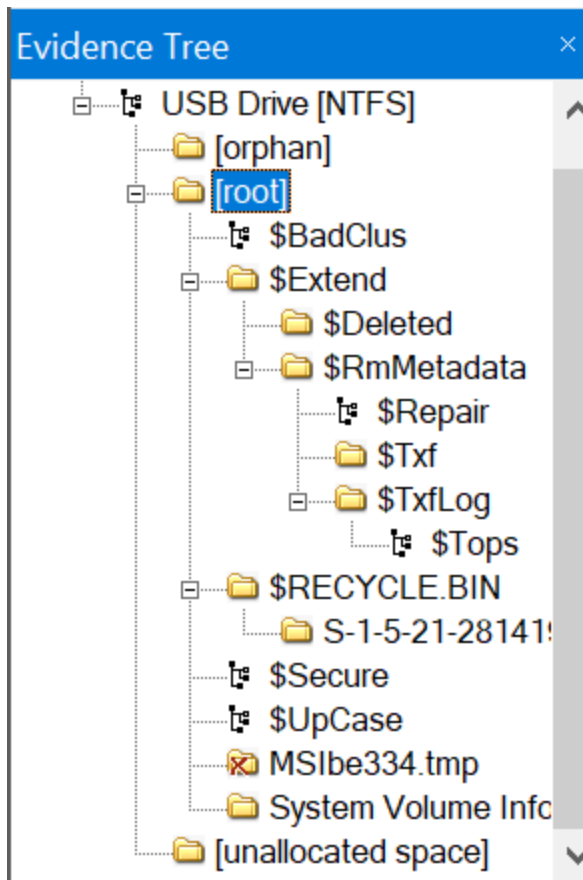
In the Select Source dialog, click Image File and click Next.



Then browse to the image file you created.



In the Evidence Tree pane, expand the tree to see the [root] folder to view the files on the imaged drive.



What is the hash of the image?

The hash value of the image USB-FTK-U:

- MD5 - 1af419c7daa1bb149efcd5cdcdb05339
- SHA1 – 72c7768243308f4a83caf1031fe91f70c2eba074

What files are present in the image?

There are different files and folders available under the root directory. In \$Deleted folder there is a I30 file slack file.

Name	Size	Type	Date Modified
\$I30	0	NTFS Index Allocation	8/2/2019 7:52:58 PM
\$I30.FileSlack	64	File Slack	

There are file types of xls, and desktop.ini in the \$Recycle.bin folder. There are xls, doc, docx, jpg, png and one executable files available in root directory.

USB-FTKE01	Name	Size	Type	Date Modified
USB Drive [NTFS]				
[orphan]				
[root]				
\$BadClus				
\$Extend				
\$Deleted				
\$RmMetadata				
\$RECYCLE.BIN				
S-1-5-21-28141984				
\$Secure				
\$UpCase				
MSIbe334.tmp				
	\$IAH3DU9.xls	1	Regular File	8/5/2019 10:18:39 PM
	desktop.ini	1	Regular File	8/2/2019 7:54:56 PM

\$Extend	1	Directory	8/2/2019...
\$RECYCLE.BIN	1	Directory	8/2/2019...
MSIbe334.tmp	1	Directory	9/1/2021...
System Volum...	1	Directory	8/2/2019...
\$AttrDef	3	Regular F...	8/2/2019...
\$BadClus	0	Regular F...	8/2/2019...
\$Bitmap	128	Regular F...	8/2/2019...
\$Boot	8	Regular F...	8/2/2019...
\$I30	4	NTFS Ind...	9/1/2021...
\$LogFile	8,832	Regular F...	8/2/2019...
\$MFT	256	Regular F...	8/2/2019...
\$MFTMirr	4	Regular F...	8/2/2019...
\$Secure	1	Regular F...	8/2/2019...
\$TXF_DATA	1	NTFS Log...	9/1/2021...
\$UpCase	128	Regular F...	8/2/2019...
\$Volume	0	Regular F...	8/2/2019...
20160425_14...	2,587	Regular F...	7/22/201...
Annual Payroll...	32	Regular F...	7/8/2015...
Balance Sheet...	30	Regular F...	7/8/2015...
Co Emp.xls	27	Regular F...	7/8/2015...
DCP_1255.jpg	266	Regular F...	9/21/200...
Employer List...	33	Regular F...	7/8/2015...
Images Profit...	47	Regular F...	7/8/2015...
IMG_2016011...	901	Regular F...	12/12/20...
IMG 2016011...	1,104	Regular F...	12/12/20...

 Employer List...	33	Regular F...	7/8/2015...
 Images Profit...	47	Regular F...	7/8/2015...
 IMG_2016011...	901	Regular F...	12/12/20...
 IMG_2016011...	1,104	Regular F...	12/12/20...
 LairNetPutty.exe	504	Regular F...	4/27/201...
 Online.docx	17	Regular F...	7/8/2015...
 Profit Potentia...	28	Regular F...	7/8/2015...
 Qtr 1 Emp.xls	23	Regular F...	7/8/2015...
 Rocky Mounta...	23	Regular F...	7/8/2015...
 Screenshot_20...	157	Regular F...	5/22/201...
 Stock Club.xls	71	Regular F...	7/8/2015...
 Summary.xls	36	Regular F...	7/8/2015...

Other than root, orphan and unallocated directories are existed. In Orphan folder there are no files or folders existed. In unallocated files there are different type of files available and file type is unallocated space.

001941	12	Unallocated Space
002031	4	Unallocated Space
005720	102,400	Unallocated Space
031320	102,400	Unallocated Space
056920	102,400	Unallocated Space
082520	102,400	Unallocated Space
1004608	102,400	Unallocated Space
1030208	70,396	Unallocated Space
108120	102,400	Unallocated Space
133720	102,400	Unallocated Space
159320	102,400	Unallocated Space
184920	102,400	Unallocated Space
210520	102,400	Unallocated Space
236120	95,132	Unallocated Space
262143	4	Unallocated Space
262208	102,400	Unallocated Space
287808	102,400	Unallocated Space
313408	102,400	Unallocated Space
339008	102,400	Unallocated Space
364608	102,400	Unallocated Space
390208	102,400	Unallocated Space
415808	102,400	Unallocated Space
441408	102,400	Unallocated Space
467008	102,400	Unallocated Space
492608	102,400	Unallocated Space
518208	102,400	Unallocated Space

What is the hash of the image USB-FTK-F?

The hash values for the image USB-FTK-F are:

- MD5 - cf302355dbff780e924dc247cc63e6da
- SHA1 – 94cbd4f7d63d0bde686ba2f3b4d8ed4eabee55fd

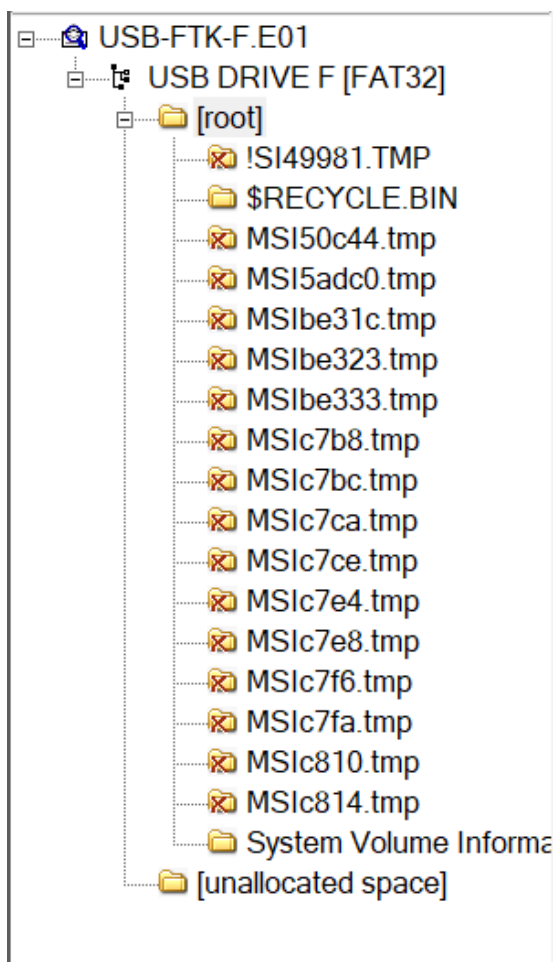
What files are present in the image?

In the USB-F: Drive image there are different folders and files are existed under the root. Mostly, .tmp folders are existed with the red X mark and the space available in those folders are zero. In the \$RECYCLE.BIN folder .ini file is there.

You can observe from the below screenshots of those files and folders.

Name	Size	Type	Date Mo...	
 ISI49981.TMP	4	Directory	8/31/202...	
 \$RECYCLE.BIN	4	Directory	8/5/2019...	
 MSI50c44.tmp	4	Directory	8/31/202...	
 MSI5adc0.tmp	4	Directory	8/11/201...	
 MSIbe31c.tmp	4	Directory	8/31/202...	
 MSIbe323.tmp	4	Directory	8/31/202...	
 MSIbe333.tmp	4	Directory	8/31/202...	
 MSIC7b8.tmp	4	Directory	8/31/202...	
 MSIC7bc.tmp	4	Directory	8/31/202...	
 MSIC7ca.tmp	4	Directory	8/31/202...	
 MSIC7ce.tmp	4	Directory	8/31/202...	
 MSIC7e4.tmp	4	Directory	8/31/202...	
 MSIC7e8.tmp	4	Directory	8/31/202...	
 MSIC7f6.tmp	4	Directory	8/31/202...	
 MSIC7fa.tmp	4	Directory	8/31/202...	
 MSIC810.tmp	4	Directory	8/31/202...	
 MSIC814.tmp	4	Directory	8/31/202...	
 System Volum...	4	Directory	8/5/2019...	
 20160425_14...	2,587	Regular F...	7/22/201...	
 Annual Payroll...	32	Regular F...	7/7/2015...	
 Annual Payroll...	32	Regular F...	7/7/2015...	
 Annual Payroll...	32	Regular F...	7/7/2015...	
 Balance Sheet....	30	Regular F...	7/7/2015...	
 Co Emp.xls	27	Regular F...	7/7/2015...	
 DCP 1255.JPG	266	Regular F...	9/20/200...	

 DCP_1255.JPG	266	Regular F...	9/20/200...
 Employer List...	33	Regular F...	7/7/2015...
 Images Profit...	47	Regular F...	7/7/2015...
 IMG_2016011...	901	Regular F...	12/12/20...
 IMG_2016011...	1,104	Regular F...	12/12/20...
 LairNetPutty.exe	504	Regular F...	4/26/201...
 Online.docx	17	Regular F...	7/7/2015...
 Profit Potentia...	28	Regular F...	7/7/2015...
 Qtr 1 Emp.xls	23	Regular F...	7/7/2015...
 Rocky Mounta...	23	Regular F...	7/7/2015...
 Screenshot_20...	157	Regular F...	5/22/201...
 Stock Club.xls	71	Regular F...	7/7/2015...
 Summary.xls	36	Regular F...	7/7/2015...



In the system volume information there are index, and wps files are existed. Remaining files are jpg, png, doc, docx, xsl, and one executable file are existed under the root.

Name	Size	Type	Date Mo...
IndexerVolum...	1	Regular F...	8/5/2019...
WPSettings.dat	1	Regular F...	8/11/201...
WPSettings.da...	4	File Slack	

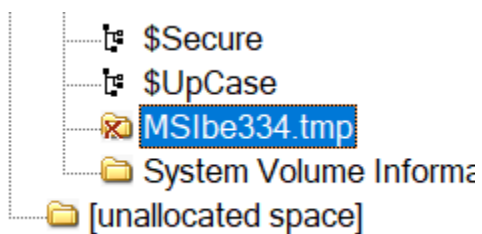
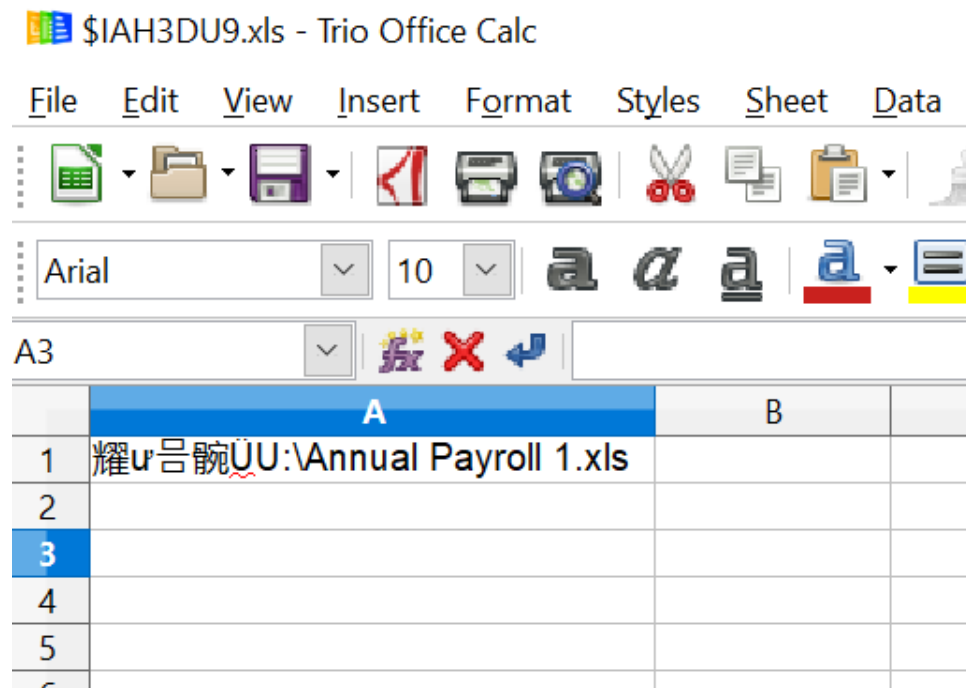
Do you see any files that show a red X? If so, can you recover them?

Drive U:

There is one file(\$IAH3DU9.xls) and one folder (MSIbe334.tmp). We can recover them by exporting files to our computer. The \$IAH3DU9.xls files consist of data and below screenshot shows the data. The Folders doesn't have any information or files existed.

Name	Size	Type	Date Modified
\$IAH3DU9.xls	1	Regular File	8/5/2019 10:18:39 PM
desktop.ini	1	Regular File	8/2/2019 7:54:56 PM

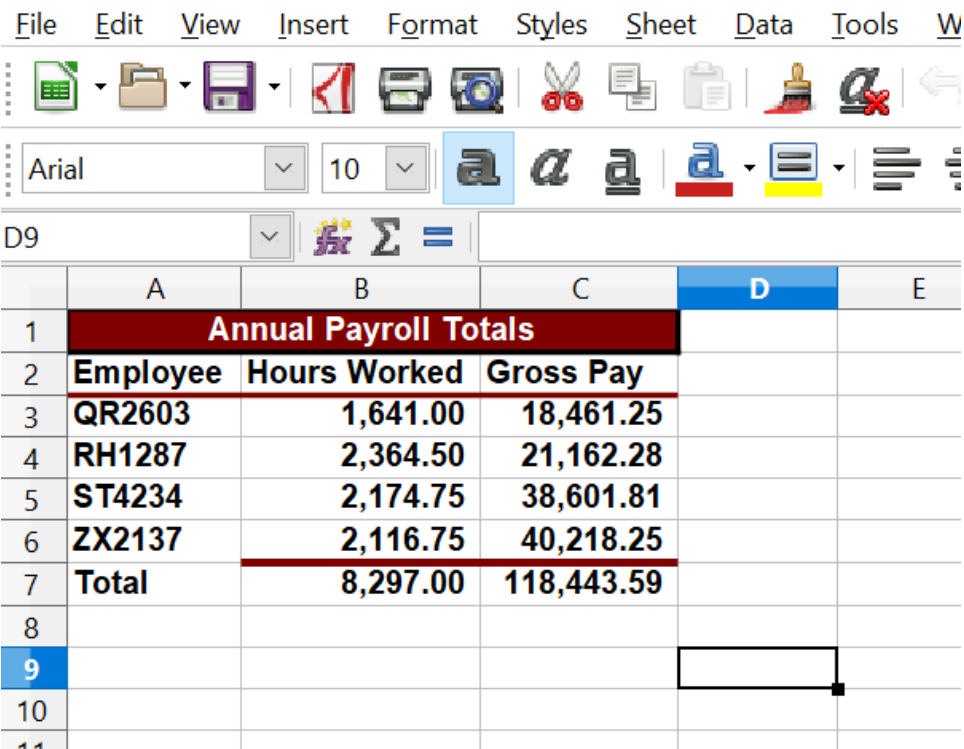
The below figure shows the content available in \$IAH3DU9.xls.



Drive F:

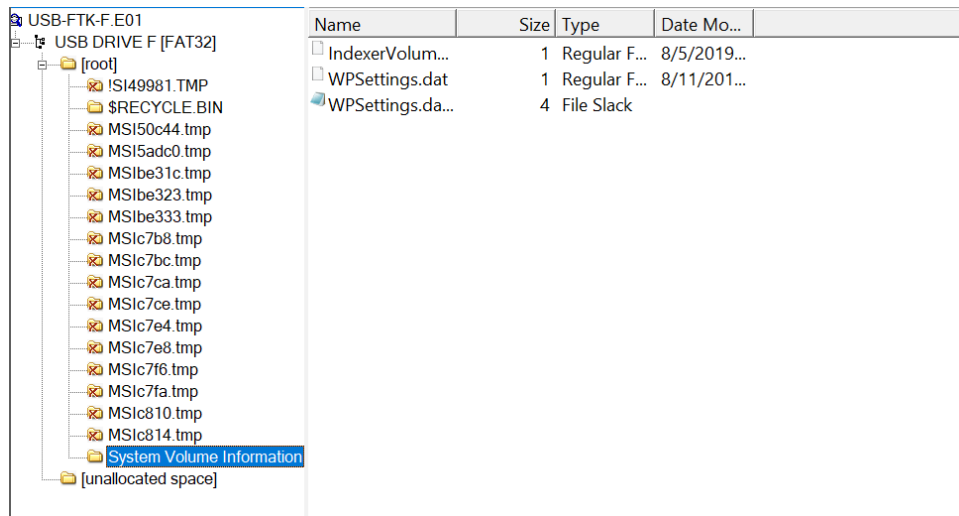
Mostly, .tmp folders are existed with the red X mark and the space available in those folders are zero. With the red X mark there is are few files and when I export to the computer and there are xls file and in that file, employee and their salary details are available.

Name	Size	Type	Date Mo...	
 !SI49981.TMP	4	Directory	8/31/202...	
 \$RECYCLE.BIN	4	Directory	8/5/2019...	
 MSI50c44.tmp	4	Directory	8/31/202...	
 MSI5adc0.tmp	4	Directory	8/11/201...	
 MSIbe31c.tmp	4	Directory	8/31/202...	
 MSIbe323.tmp	4	Directory	8/31/202...	
 MSIbe333.tmp	4	Directory	8/31/202...	
 MSIs7b8.tmp	4	Directory	8/31/202...	
 MSIs7bc.tmp	4	Directory	8/31/202...	
 MSIs7ca.tmp	4	Directory	8/31/202...	
 MSIs7ce.tmp	4	Directory	8/31/202...	
 MSIs7e4.tmp	4	Directory	8/31/202...	
 MSIs7e8.tmp	4	Directory	8/31/202...	
 MSIs7f6.tmp	4	Directory	8/31/202...	
 MSIs7fa.tmp	4	Directory	8/31/202...	
 MSIs810.tmp	4	Directory	8/31/202...	
 MSIs814.tmp	4	Directory	8/31/202...	
 System Volum...	4	Directory	8/5/2019...	
 20160425_14...	2,587	Regular F...	7/22/201...	
 Annual Payroll...	32	Regular F...	7/7/2015...	
 Annual Payroll...	32	Regular F...	7/7/2015...	
 Annual Payroll...	32	Regular F...	7/7/2015...	
 Balance Sheet....	30	Regular F...	7/7/2015...	
 Co Emp.xls	27	Regular F...	7/7/2015...	
 DCP 1255.JPG	266	Regular F...	9/20/200...	



Are the file system contents of each image the same?

Tool	Pro Discover And FTK Image
Drive U:	The file system content is same and the files present inside the root are same.
Drive F:	The file system contents are same and there are .tmp files in both the images with the file size zero. System Volume information in the both the images are same.

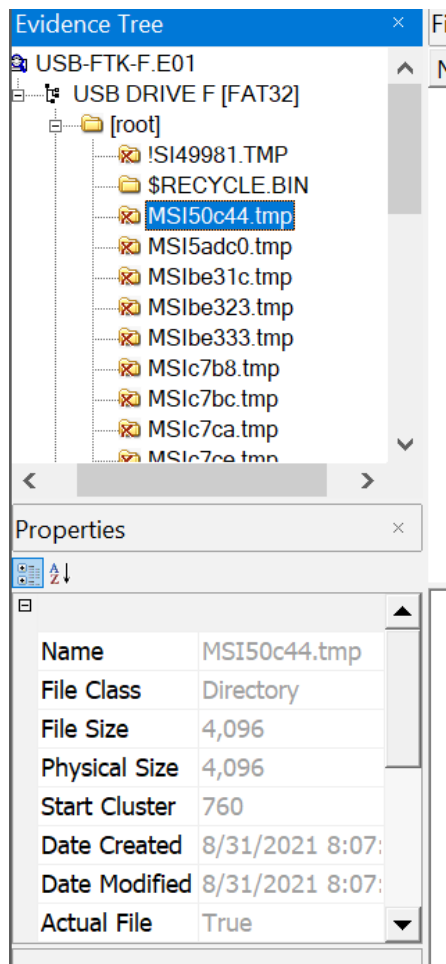


Are there any files missing from an image?

A) No, all the files are available in both the images.

Which tool would you prefer to use as your main tool, and why? (There is no right or wrong answer)

A) For converting disk to image, I will prefer FTK imager tool because it's going to give you directly the hash values, whereas in Pro Discover tool we need to use Windows PowerShell to get the hash values. For analyzing the evidence, FTK imager will load the image faster while compared to Pro Discover and I feel FTK imager user interface is great, you can export the files directly to your computer without copying them. At the left bottom of the tool, you can directly see the properties of selected files.



Did you notice any differences between the images of the two USB drives (F: and U:)? If so, what differences?

A) USB Drive U: The Drive U: uses the NTFS (The current file system that Windows likes to use by default is NTFS. When you install Windows, it uses the NTFS file system to format your hard drive. You won't run into NTFS file and partition size limits because they are theoretically so large.) file system

USB Drive F: It uses FAT32 file system and you can see that in the below figure. In Drive F: you can find the so many .tmp files while compared to Drive U: I think these are the main differences that I found of.

.tmp Files:

Name	Size	Type	Date Mo...	
❌ ISI49981.TMP	4	Directory	8/31/202...	
📁 \$RECYCLE.BIN	4	Directory	8/5/2019...	
❌ MSI50c44.tmp	4	Directory	8/31/202...	
❌ MSI5adc0.tmp	4	Directory	8/11/201...	
❌ MSIbe31c.tmp	4	Directory	8/31/202...	
❌ MSIbe323.tmp	4	Directory	8/31/202...	
❌ MSIbe333.tmp	4	Directory	8/31/202...	
❌ MSIs7b8.tmp	4	Directory	8/31/202...	
❌ MSIs7bc.tmp	4	Directory	8/31/202...	
❌ MSIs7ca.tmp	4	Directory	8/31/202...	
❌ MSIs7ce.tmp	4	Directory	8/31/202...	
❌ MSIs7e4.tmp	4	Directory	8/31/202...	
❌ MSIs7e8.tmp	4	Directory	8/31/202...	
❌ MSIs7f6.tmp	4	Directory	8/31/202...	
❌ MSIs7fa.tmp	4	Directory	8/31/202...	
❌ MSIs810.tmp	4	Directory	8/31/202...	
❌ MSIs814.tmp	4	Directory	8/31/202...	
📁 System Volum...	4	Directory	8/5/2019...	
📄 20160425_14...	2,587	Regular F...	7/22/201...	
❌ Annual Payroll...	32	Regular F...	7/7/2015...	
❌ Annual Payroll...	32	Regular F...	7/7/2015...	
❌ Annual Payroll...	32	Regular F...	7/7/2015...	
📊 Balance Sheet....	30	Regular F...	7/7/2015...	
📊 Co Emp.xls	27	Regular F...	7/7/2015...	
📄 DCP 1255.JPG	266	Regular F...	9/20/200...	

Exporting files:

Name	Size	Type	Date Modified
📄 \$IAH3DU9.x...			8/5/2019 10:18:39 PM
📄 desktop.ini			8/2/2019 7:54:56 PM

☐ Export Files...
☐ Export File Hash List...
☐ Add to Custom Content Image (AD1)

Conclusion:

In this lab, ProDiscover and FTK Imager were used to analyze forensic images of two USB drives formatted with different file systems. Both tools revealed consistent file system contents across images. The analysis demonstrated how deleted files, system files, and unallocated space can be examined and recovered. Overall, this lab provided practical experience in disk image analysis, file recovery, and understanding differences between NTFS and FAT32 file systems.

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